Employment Onsite Opportunity

This documentation outlines the end-to-end data engineering lifecycle for the **Employment Opportunity Analytics** project. It tracks the data from its raw state in AWS through a multi-cloud transformation pipeline, ending in an executive Power BI dashboard.

Stage	Platform	Primary Tool/Service
Ingestion	AWS	S3
Orchestration	GCP	Cloud Composer (Airflow)
Transformation	Snowflake	SQL, Stages, Storage Integrations
Storage	GCP	GCS, BigQuery
Visualization	Microsoft	Power BI Web

1. Data Ingestion (AWS S3)

The lifecycle begins with raw data residency in Amazon S3.

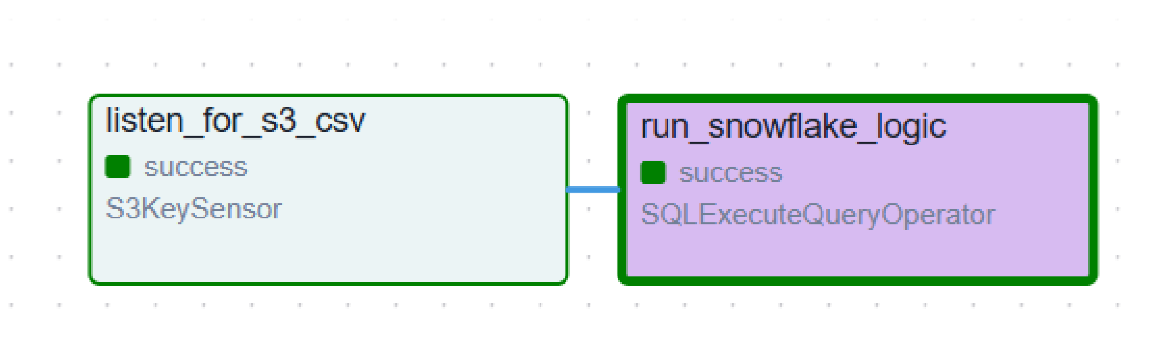
- **Source:** A CSV file named `employment_onsite_opportunity.csv` is uploaded to the S3 bucket `mock-proj-prateek-khandekar`.
- **Automation:** An Airflow `S3KeySensor` is configured to "listen" for this specific file, ensuring the pipeline only triggers once the data is physically present.

2. Multi-Cloud Orchestration (Apache Airflow / Cloud Composer)

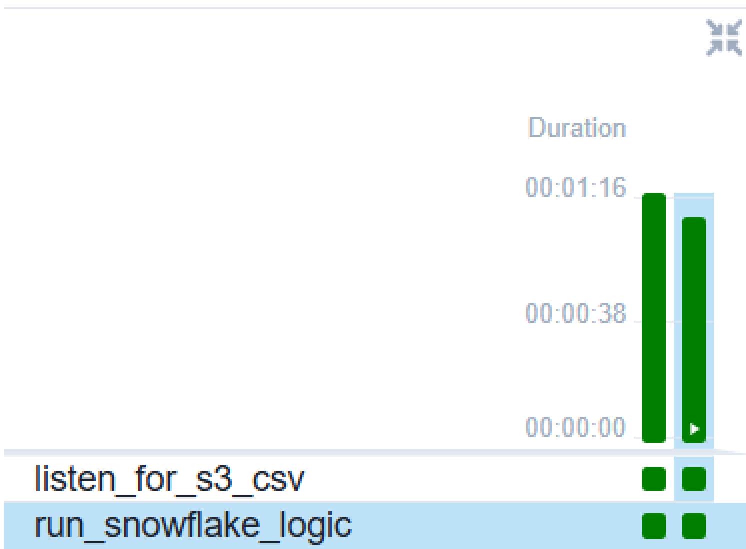
Google Cloud Composer (managed Airflow) acts as the "brain" of the operation.

- **Provider Packages:** Installed `apache-airflow-providers-snowflake` and `apache-airflow-providers-amazon` to bridge the gap between AWS and Snowflake.
- **Connection Management:** Securely stored AWS Access Keys and Snowflake credentials within the Airflow Connection URI to avoid hardcoding secrets.
- **Task Flow:** Orchestrated a DAG that sequences the data movement: **S3 Sensor → Snowflake SQL Execution**.

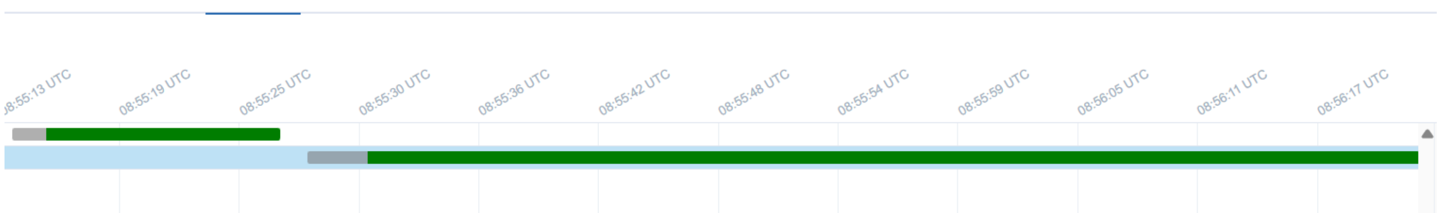
Charts for Apache airflow



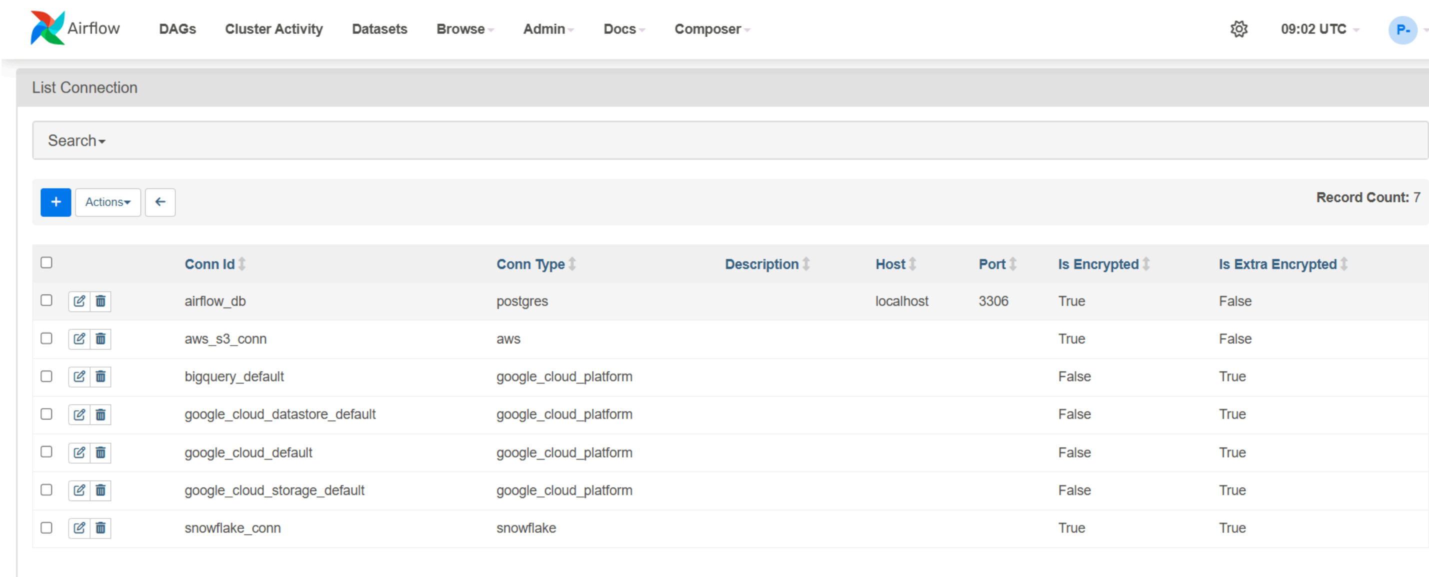
Success status indicator



Gantt charts

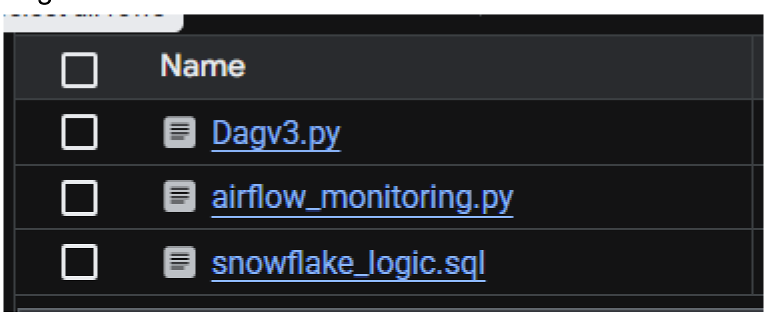


Connections in the airflow UI (see the snowflake_conn and aws_s3_conn which connects airflow to both these services)



Conn Id	Conn Type	Description	Host	Port	Is Encrypted	Is Extra Encrypted
airflow_db	postgres		localhost	3306	True	False
aws_s3_conn	aws				True	False
bigquery_default	google_cloud_platform				False	True
google_cloud_datastore_default	google_cloud_platform				False	True
google_cloud_default	google_cloud_platform				False	True
google_cloud_storage_default	google_cloud_platform				False	True
snowflake_conn	snowflake				True	True

Dags bucket



3. Data Transformation & Cleaning (Snowflake)

Data is ingested into Snowflake for heavy-duty processing.

- **Ingestion:** Used the `COPY INTO` command to pull the CSV from the S3 External Stage into a raw landing table.
- **Data Cleaning:** Performed schema enforcement, handled null values and filtered out inconsistent records

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4. Egress to Google Cloud (Snowflake to GCS)

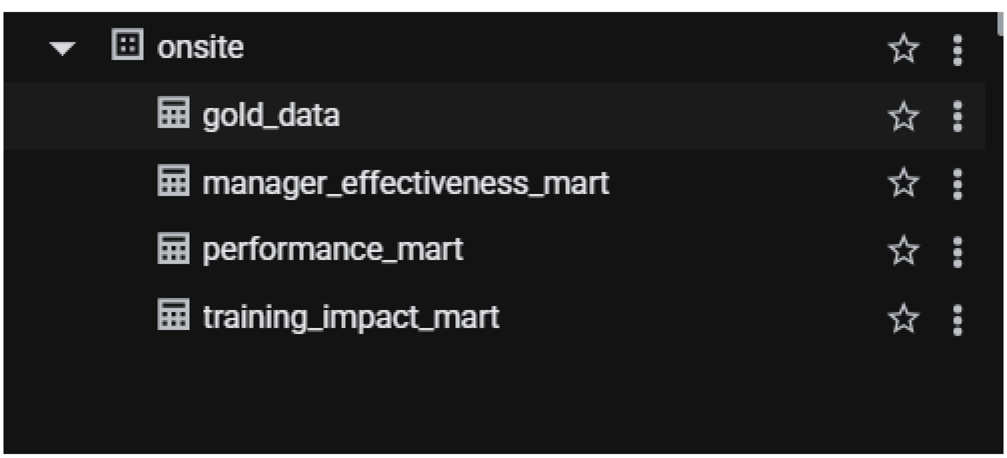
Once cleaned, the data must return to the Google ecosystem.

- **Storage Integration:** Configured a Snowflake Storage Integration to allow secure write access to Google Cloud Storage (GCS).
- **Unloading:** Executed a `COPY INTO @gcs_stage` command to export the transformed data as a compressed CSV into a GCS bucket.

5. Data Mart Creation (BigQuery)

The data is finalized in BigQuery to serve as the "Single Source of Truth."

- **Schema Autodetection:** Used the `GCSToBigQueryOperator` to load the exported file into the `onsite` dataset.
- Added the table from gcs to bigquery using the create table feature via the UI .
- **Mart Logic:** Finalized the `training_impact_mart` by categorizing employees into "High Impact" or "Positive Impact" based on their `PHASE` and salary thresholds.

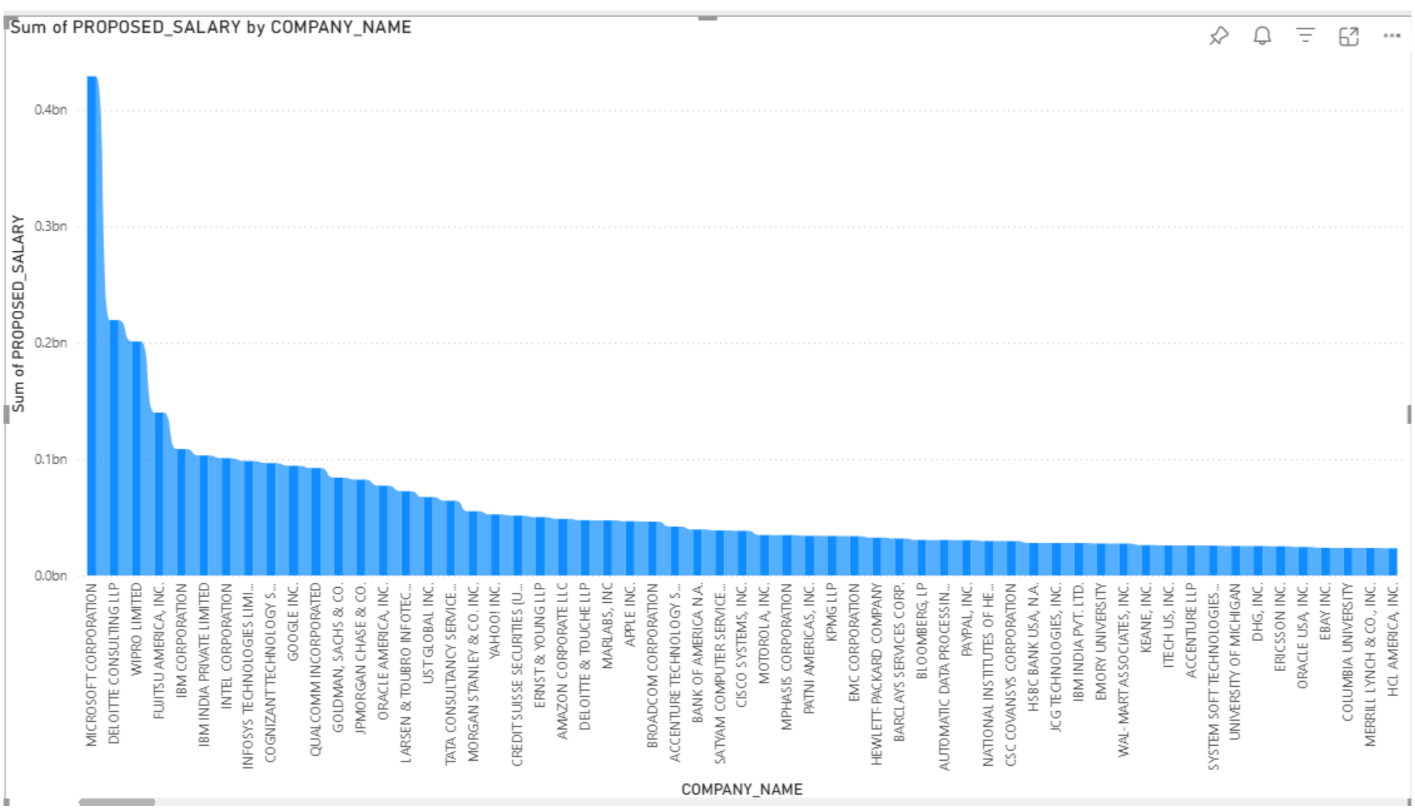


6. Data Visualization (Power BI)

The "Frontend" layer translates the data mart into actionable insights.

- **Connectivity:** Established a direct connection between Power BI and BigQuery by creating a GCP service account.
- **DAX Analytics:** Developed custom measures for:
  - Sum of proposed Salary by company name
  - Effective Success Rate
- **Executive Dashboarding:** Built visuals for Manager Effectiveness, Training Impact on Promotions, and Departmental Resignation rates.
- **Alerting:** Configured Data Alerts to notify stakeholders via email if Cloud Ops errors are detected in the pipeline.

Sum of proposed Salary by company name

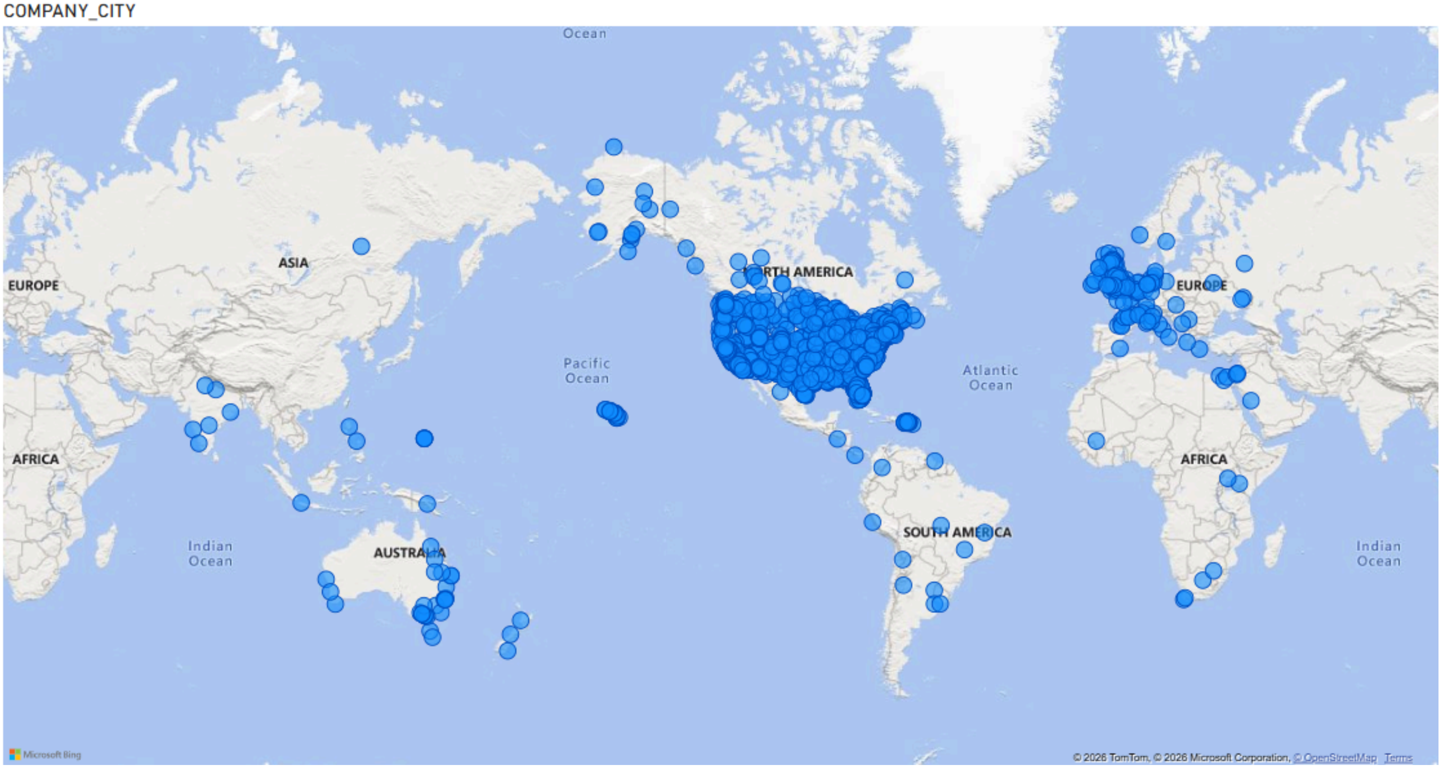


DAX example

```
Effective Success Rate =
DIVIDE(
    SUMX('Manager_Effectiveness_Mart', 'Manager_Effectiveness_Mart'[TOTAL_APPLICATIONS] *
    'Manager_Effectiveness_Mart'[SUCCESS_RATE_PERCENTAGE]),
    SUM('Manager_Effectiveness_Mart'[TOTAL_APPLICATIONS])
)
```

Employee vs Country Map

COMPANY_CITY



7. IAM Roles

Type	Principal	Name	Role	Security insights
	97281595932-compute@developer.gserviceaccount.com	Compute Engine default service account	BigQuery Data Editor	Advanced security insight
			BigQuery Job User	Advanced security insight
			Composer Worker	Advanced security insight
			Editor	
			Logs Writer	Advanced security insight
			Pub/Sub Publisher	Advanced security insight
			Service Account User	Advanced security insight
			Storage Object Viewer	Advanced security insight
	kpxvnrnrln@awsapsouth1-1-153a.iam.gserviceaccount.com		Storage Object Admin	Advanced security insight
	powerbi-user-718@mock-proj-488913.iam.gserviceaccount.com	powerbi-user	BigQuery Data Viewer	Advanced security insight
			BigQuery Job User	Advanced security insight
			BigQuery Read Session User	Advanced security insight
	prateekkhandekar01@gmail.com	Prateek	Advisory Notifications Viewer	Advanced security insight
			Organization Administrator	Advanced security insights
			Owner	
			Project Mover	Advanced security insight
			Service Usage Admin	Advanced security insight